(FP2) 4. Integration 专题作业2

NO. 1

OR

The polar equation of a curve C is $r = a(1 + \cos\theta)$ for $0 \leq \theta < 2\pi$, where a is a positive constant.

i. Sketch C. [2]

ii. Show that the cartesian equation of C is [2]

$$x^2 + y^2 = a\left(x + \sqrt{(x^2 + y^2)}\right).$$

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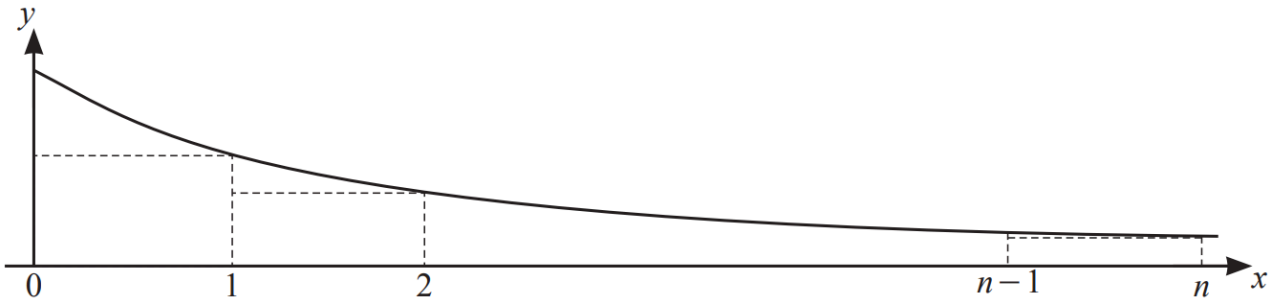
iii. Find the area of the sector of C between $\theta = 0$ and $\theta = \frac{1}{3}\pi$. [4]

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iv. Find the arc length of C between the point where $\theta = 0$ and the point where $\theta = \frac{1}{3}\pi$. [5]

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NO. 2



The diagram shows the curve $y = \frac{1}{\sqrt{x^2+x+1}}$ for $x \geq 0$, together with a set of n rectangles of unit width. By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^{n} \frac{1}{\sqrt{r^2+r+1}} < \ln\left(\tfrac{1}{3}+\tfrac{2}{3}n+\tfrac{2}{3}\sqrt{n^2+n+1}\right).$$ [10]

NO. 3

a. Sketch the graph of $y = \coth x$ for $x > 0$ and state the equations of the asymptotes. [2]

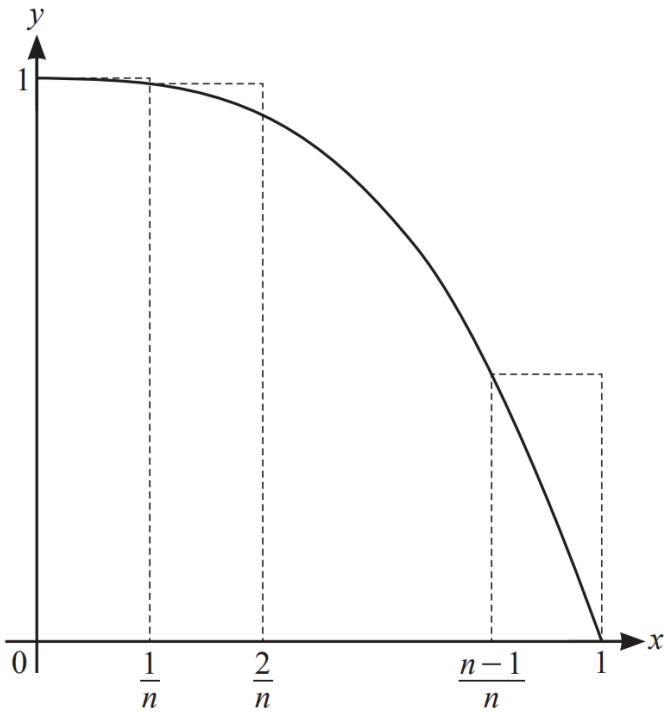
b. Starting from the definitions of coth and cosech in terms of exponentials, prove that $\coth^2 x - \operatorname{cosech}^2 x = 1$. [3]

The curve C has equation $y = \ln \coth(\frac{1}{2}x)$ for $x > 0$.

c. Show that $\frac{dy}{dx} = -\operatorname{cosech} x$. [3]

d. It is given that the arc length of C from $x = a$ to $x = 2a$ is $\ln 4$, where a is a positive constant.
Show that $\cosh a = 2$ and find, in logarithmic form, the exact value of a. [7]

NO. 4



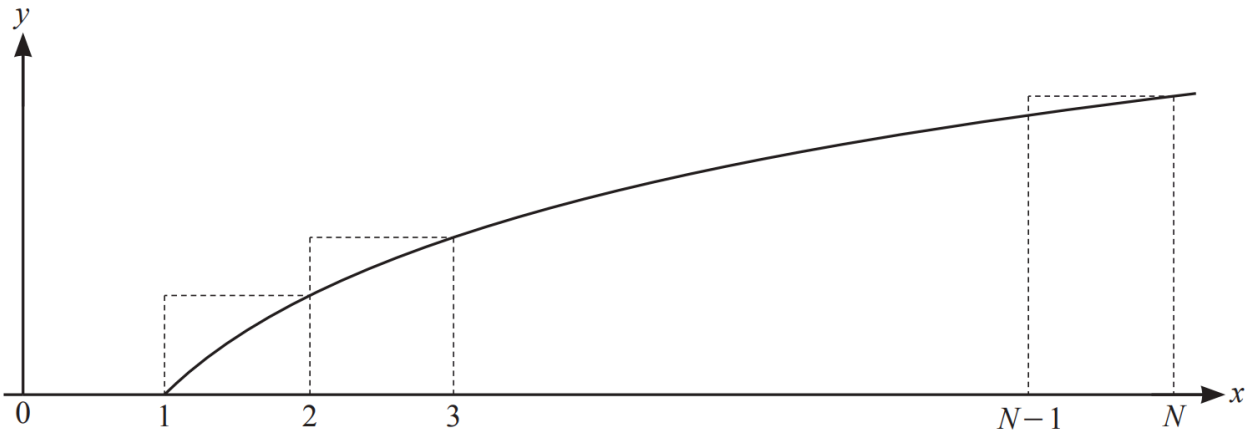
The diagram shows the curve with equation $y = 1 - x^3$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

a. By considering the sum of the areas of the rectangles, show that

$$\int_0^1 (1-x^3)\,dx \leqslant \frac{3n^2+2n-1}{4n^2}.$$ [4]

b. Use a similar method to find, in terms of n, a lower bound for $\int_0^1 (1-x^3)\,dx$. [4]

NO. 5



The diagram shows the curve with equation $y = \ln x$ for $x \geq 1$, together with a set of $(N - 1)$ rectangles of unit width.

a. By considering the sum of the areas of these rectangles, show that $\ln N! > N \ln N - N + 1$. [5]

b. Use a similar method to find, in terms of N, an upper bound for $\ln N!$. [3]

NO. 6

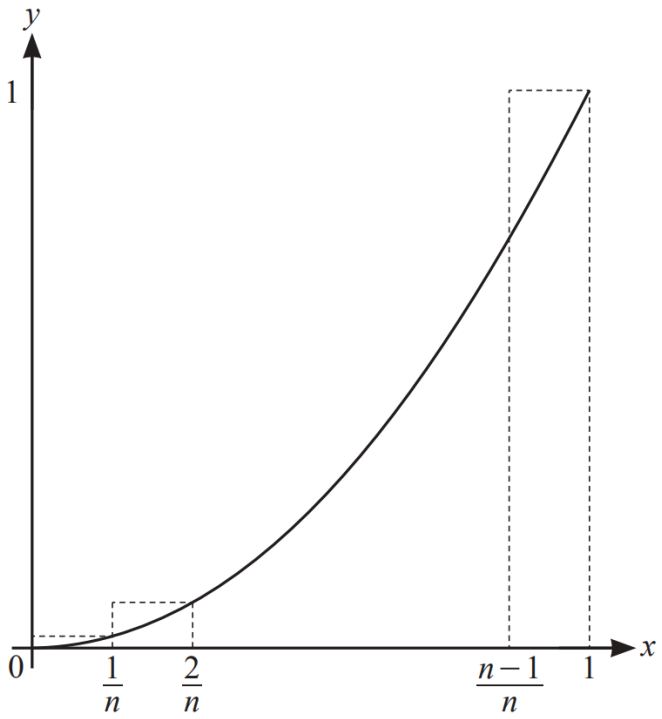
The curves C_1 : y =cosh x and C_2 : y = sinh 2x intersect at the point where x = a.

a. Find the exact value of a, giving your answer in logarithmic form. [4]

b. Sketch C_1 and C_2 on the same diagram. [2]

c. Find the exact value of the length of the arc of C_1 from x = 0 to x = a. [5]

NO. 7



The diagram shows the curve with equation $y = x^2$ for $0 \le x \le 1$, together with a set of n rectangles of width $\frac{1}{n}$.

a. By considering the sum of the areas of these rectangles, show that

$$\int_0^1 x^2 \, dx < \frac{2n^2 + 3n + 1}{6n^2}.$$ [4]

b. Use a similar method to find, in terms of n, a lower bound for $\int_0^1 x^2 \, dx$. [4]

NO. 8

The curve C has parametric equations

$$x = \frac{1}{2}e^{2t} - \frac{1}{3}t^3 - \frac{1}{2}, \qquad y = 2e^t(t-1), \qquad \text{for } 0 \leqslant t \leqslant 1.$$

Find the exact length of C. [7]

NO. 9

A curve has equation $y = e^x$ for $\ln\frac{4}{3} \leqslant x \leqslant \ln\frac{12}{5}$. The area of the surface generated when the curve is rotated through 2π radians about the x-axis is denoted by A.

a. Use the substitution $u = e^x$ to show that $A = 2\pi\int_{\frac{4}{3}}^{\frac{12}{5}} \sqrt{1+u^2}\,\mathrm{d}u$. [2]

b. Use the substitution $u = \sinh v$ to show that $A = \pi\left(\frac{904}{225} + \ln\frac{5}{3}\right)$. [6]